ResearchGraph

Multi-Agent Research Discovery Platform

Backend disconnected | Showing demo graph | No active job | Retry connection

Research papers

5 demo · 0 uploaded

0 Uploaded	0 Processed	0 Failed

Drop 30–35 PDFs or browse

PDF only · multiple files at once

Uploads are off until `POST /papers/upload` responds. Nothing is marked uploaded until the server confirms.

DEMO PAPERS — SYNTHETIC

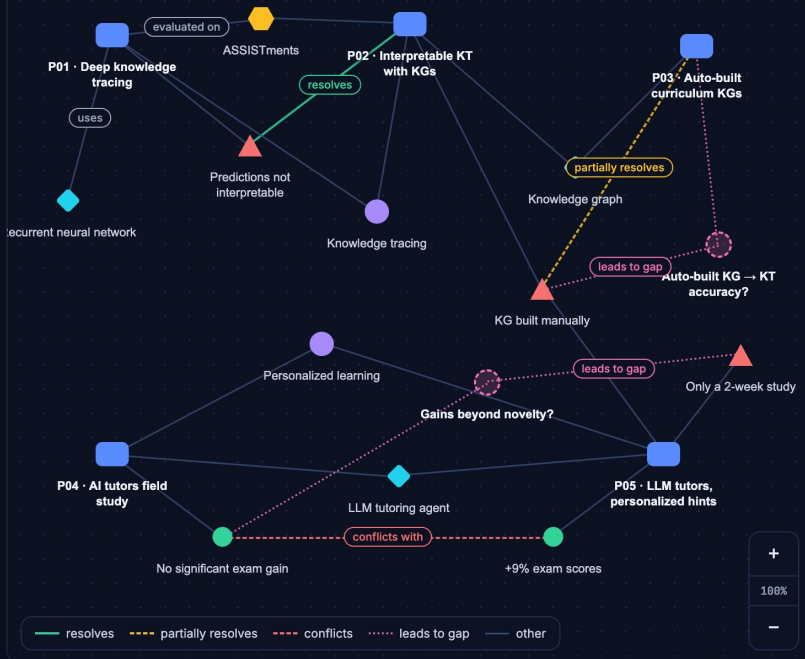
- P01 Deep Knowledge Tracing for Personalized Learning in Online Courses
  2021 · demo · not processed by backend
- P02 Interpretable Knowledge Tracing with Knowledge Graphs
  2022 · demo · not processed by backend
- P03 Automatic Curriculum Knowledge Graph Construction with LLMs
  2023 · demo · not processed by backend
- P04 AI Tutoring Agents in Real Classrooms: A Field Study
  2023 · demo · not processed by backend
- P05 LLM Tutors Improve Learning Outcomes with Personalized Hints
  2024 · demo · not processed by backend

Search nodes...

Papers 5 | Concepts 2 | Methods 3 | Datasets 1 | Findings 2 | Limitations 3 | Gaps 2

Fit to screen | Reset view

DEMO GRAPH · SYNTHETIC DATA



+ 100% −

Evidence inspector

Nothing selected

Click a node or a relationship label in the graph to see its evidence, papers and page references.

Research discovery

DEMO INSIGHTS — Generated from the 5 synthetic papers. Every item is a hypothesis to check, not a finding.

Common research themes

- Knowledge tracing for personalized learning
  P01 · P02 · P05
- Knowledge graphs for interpretability
  P02 · P03 · P05

Similar methodologies

- Graph-grounded student models
  P02 · P03 · P05
- Classroom trials with a control group
  P04 · P05

Potentially conflicting findings

- Exam gains from AI tutors: none (P04) vs +9% (P05)
  F1 ↔ F2

Underexplored areas

- Auto-built KGs never tested for knowledge-tracing accuracy
  P03 · L2
- Learning gains measured past two weeks
  P05 · L3

Suggested directions

- Hypothesis: swap P05's manual KG for P03's auto-built KG and run a semester-long trial
  G1 + G2

Processing batch

`job_[ID]` EXAMPLE STATE

Polling `GET /jobs/{job_id}` every few seconds · safe to leave this page

34	32	18 / 31	3
Files selected	Uploaded (server confirmed)	Extracted by LLM	Need attention

Pipeline

Inference runs on the HPC cluster · elapsed [mm:ss]

✓	✓	✓	4	5	6	7	8	9	10
Batch ingestion	Parallel text extraction	Section-aware chunking	LLM structured extraction	Entity normalization	Cross-paper relationships	Graph construction	Gap identification	Evidence validation	Visualization
32 of 34 accepted	31 ok · 1 failed	31 papers	18 / 31 papers	waiting	waiting	waiting	waiting	waiting	waiting

Files

All 34 | Processing 13 | Done 18 | Needs attention 3

Add more PDFs | Cancel job

FILE	SIZE	UPLOAD	PROCESSING	PAGES	MESSAGE
paper_01.pdf	[size]	Uploaded	Done	[n]	[n] findings · [n] limitations
paper_02.pdf	[size]	Uploaded	Done	[n]	[n] findings · [n] limitations
paper_03.pdf	[size]	Uploaded	LLM extraction	[n]	Sent to HPC queue
paper_04.pdf	[size]	Uploaded	LLM extraction	[n]	Sent to HPC queue
paper_05.pdf	[size]	Uploaded	Queued	[n]	—
paper_06.pdf	[size]	Uploaded	Queued	[n]	—
scanned_paper.pdf	[size]	Uploaded	Failed	[n]	No text layer found (scanned PDF). OCR is not enabled.
notes.docx	[size]	Rejected	Skipped	—	Not a PDF. Only .pdf files are accepted.
paper_02 (1).pdf	[size]	Rejected	Skipped	—	Duplicate of paper_02.pdf (same file hash).
paper_07.pdf	[size]	Uploaded	Queued	[n]	—
paper_08.pdf	[size]	Uploaded	Queued	[n]	—
paper_09.pdf	[size]	Uploaded	Done	[n]	[n] findings · [n] limitations

Showing 12 of 34 · File names, sizes and page counts are placeholders.

Interface states

Every panel shows what is really true: no fake uploads, no invented evidence, demo data always labelled.

EMPTY · NO PAPERS YET

Your graph is empty

Upload 30–35 papers from one domain to map their methods, findings and gaps.

Upload PDFs | View demo graph

LOADING · JOB IN PROGRESS

Building knowledge graph Stage 4 of 10

LLM structured extraction · 18 of 31 papers

The previous graph stays usable until the new one is ready.

DISCONNECTED · ENDPOINT UNAVAILABLE

Can't reach the backend. `GET /graph` did not respond. You are looking at the demo graph, not your papers.

`GET /papers`	unreachable
`GET /graph`	unreachable
`GET /insights`	404 · not implemented

Retry connection

PARTIAL UPLOAD FAILURE

32 of 34 files uploaded

Uploaded files are processing. Fix or remove these before you retry:

`notes.docx`
Not a PDF. Only .pdf files are accepted.

`paper_02 (1).pdf`
Duplicate of paper_02.pdf.

Choose replacement files | Dismiss

LARGE GRAPH · FILTERED VIEW

Showing **[N]** of **[TOTAL]** nodes: papers, gaps and their direct neighbors. Expand a node to load more.

HIDDEN BY DEFAULT

- Concepts used by one paper
- Findings

Show everything (may be slow)

INSPECTOR · NO EVIDENCE FOUND

Method

[Entity name]

No supporting quote was extracted for this entity. It appears in the paper's metadata only, so there is no page reference to show.

Quotes that can't be matched back to the PDF text are flagged as unverified, never shown as evidence.